

Some Basic Concepts

July 31, 2024

Random Experiment

Random Experiments

Sample Space

Events

Few Set Operations

Review of Set
Operations

Some Results-Set
Theoretic

Some Results-PIE

Examples-PIE

- A procedure that is carried out under certain set of conditions is called a *random experiment*
- A random experiment might contain repetitive steps
- If repeated, then it should be repeated under same set of conditions
- Each repetition is often called a *trial*

Example:

- tossing a coin three times
- throwing two dice

Sample Space

Random Experiments

Sample Space

Events

Few Set Operations

Review of Set Operations

Some Results-Set Theoretic

Some Results-PIE

Examples-PIE

Sample Points: Outcome of a random experiment is called a *sample point*.

Example:

- tossing a coin three times – (H,T,H) (Consists of three trials)
- throwing two dice – $(5,6)$ (Consists of a single trial)

Sample Space: All possible outcomes of a random experiment is called *sample space*.

Example:

- tossing a coin three times – $\{(H,H,H), (H,H,T), (H,T,H), (H,T,T), (T,H,H), (T,H,T), (T,T,H), (T,T,T)\}$
- throwing two dice – all possible distinct pairs from the set $\{1, 2, 3, 4, 5, 6\}$

Cardinality of Sample Space:

- **Finite:** tossing a coin three times
- **Countably infinite:** Tossing a coin until a head is shown
 $\mathcal{S} = \{(H), (T, H), (T, T, H), \dots\}$
- **Uncountably infinite:** Lifetime of an electronic device
 $\mathcal{S} = [0, \infty)$

Quiz: Point out the cardinality of each of the sample space for the following.

- Yearly income of a target population
- Number of particles emitted by a certain radioactive source within a specified period of time
- Drawing a card from a well-shuffled standard deck of 52 cards
- Number of traffic accidents that occur in a specified location within a certain period of time

Event: Any subset of the sample space is called a *event*.

Example:

- tossing a coin three times
 - $\{(H,T,H), (T,H,T)\} \Leftrightarrow$ the outcome of the second trial is different from the other two trials
 - $\{(H,H,H), (H,H,T), (H,T,H), (T,H,H), (T,T,T)\} \Leftrightarrow$ Even number of heads (?)
- throwing two dice
 - $\{(5,6), (6,5), (6,6)\} \Leftrightarrow$ Sum is at least 11 (Is there any other way to define the event?)
 - $\{(1,1)\} \Leftrightarrow$ Maximum is 1

Note:

- Entire sample space is an event
- $\emptyset$ is an event

Review of Set Operations

Consider the sample space $\mathcal{S} = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8\}$ and define the events A_1 , A_2 , and A_3 as follows:

$$A_1 = \{s_1, s_2, s_3\}, A_2 = \{s_2, s_3, s_4, s_5\}, A_3 = \{s_3, s_4, s_5, s_8\}.$$

- **Complement:** $A_2^c = \{s_1, s_6, s_7, s_8\}$
- **Union:** $A_3 \cup A_1^c = \{s_3, s_4, \dots, s_8\}$
- **Intersection:** $A_1 \cap A_2 = \{s_2, s_3\}$
- **DeMorgan's Law:** $(A_1 \cap A_2)^c = A_1^c \cup A_2^c$
 - $A_1 \cap A_2 = \{s_2, s_3\}$
 - $(A_1 \cap A_2)^c = \{s_1, s_4, s_5, s_6, s_7, s_8\}$
 - $A_1^c = \{s_4, s_5, s_6, s_7, s_8\}$
 - $A_2^c = \{s_1, s_6, s_7, s_8\}$
 - $A_1^c \cup A_2^c = \{s_1, s_4, s_5, s_6, s_7, s_8\}$
- **Set Difference:** $A_1 - A_2 = \{s_1\}$

Note:

- De Morgan's Law holds for any number of sets, finite or infinite
- $A_1 - A_2 = A_1 \cap A_2^c$

- **Proposition:** $\cup_j A_j = A_1 \cup (A_1^c \cap A_2) \cup (A_1^c \cap A_2^c \cap A_3) \cup \dots$

Note: The above proposition holds for countably infinite many sets which may be non-discrete

An illustration of the above proposition:

- $A_1 = \{s_1, s_2, s_3\}, A_2 = \{s_2, s_3, s_4, s_5\}, A_3 = \{s_3, s_4, s_5, s_8\}$
- $A_1^c \cap A_2 = \{s_4, s_5\}$
- $A_1^c \cap A_2^c \cap A_3 = \{s_8\}$
- $A_1 \cup A_2 \cup A_3 = \{s_1, s_2, s_3, s_4, s_5, s_8\}$

Some Results – PIE

- **Principle of Inclusion and Exclusion (PIE):**

$$|\bigcap_{i \in [n]} A_i^c| = |\mathcal{S}| - \sum |A_i| + \sum |A_i \cap A_j| - \dots + (-1)^n |A_1 \cap A_2 \cap \dots \cap A_n|$$

Proof Sketch

- Let A_i represent the set of elements satisfying property P_i
- Let x be an element that satisfies none of the properties
 - LHS counts it exactly once
 - Since $\forall i \ x \notin A_i$, x is counted only by $\mathcal{S}$
- Let x be an element that satisfies $k \leq n$ properties
- w.l.o.g. let $x \in A_1 \cap A_2 \cap \dots \cap A_k$
 - x is counted 0 times in LHS
 - x is counted in RHS following number of times

$$\binom{k}{0} - \binom{k}{1} + \binom{k}{2} - \dots + (-1)^k \binom{k}{k} = 0$$

Examples – PIE

Q1: How many permutations of the letters

$M, A, T, H, I, S, F, U, N$

are there such that none of the words MATH, IS and FUN occurs in any of the permutations?

Q2: What is the number of integral solutions of the equation

$$x_1 + x_2 + x_3 + x_4 = 18$$

such that

$$1 \leq x_1 \leq 5, -2 \leq x_2 \leq 4, 0 \leq x_3 \leq 5, 3 \leq x_4 \leq 9?$$

A *derangement* of the set $\{1, 2, 3, \dots, n\}$ is a permutation $i_1, i_2, i_3, \dots, i_n$ such that $i_1 \neq 1, i_2 \neq 2, i_3 \neq 3, \dots, i_n \neq n$.

Q3: How many derangements of the set $\{1, 2, 3, \dots, n\}$ are possible?